

Program : Diploma in Printing Technology	
Course Code : 5103C	Course Title: Screen Printing
Semester : 5	Credits: 4
Course Category: Program Elective	
Periods per week: 4 (L:4, T:0, P:0)	Periods per semester: 60

Course Objectives:

- To guide the student to the basic concepts of the screen printing
- To provide the exposure to the essentials of screen printing and to learn how it influences the real printing world.
- To understand the stencil concept to the printing process.
- To develop idea on most exciting possibilities of screen printing in the upcoming era.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Screen printing process		Basics of Printing process	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (hours)	Cognitive Level
CO1	Describe the origin of screen printing, its special benefits, applications and various materials required to achieve the screen printing process.	15	Understanding
CO2	Explain about substrates and inks used in screen printing and various ink drying methods and related facilities.	14	Understanding
CO3	Explain about on contact & off contact printing, specialized areas, image transfer and registration techniques.	15	Understanding
CO4	Describe about various printing press classifications & configurations of screen-printing.	14	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Describe the origin of screen printing, its special benefits, applications and various materials required to achieve the screen printing process.		
M1.01	History of Screening Printing,	1	Understanding
M1.02	Merits, Demerits and applications of screen printing.	1	Understanding
M1.03	Screen printing materials- Screen, Frame, Squeegee and Stencil.	3	Understanding
M1.04	Screens – types, selection, stretching, preparation & reclamation.	3	Understanding
M1.05	Trouble shooting clogged screens, Care and storage of screens.	1	Understanding
M1.06	Squeegee considerations.	3	Understanding
M1.07	Stencils-classifications and preparation.	3	Understanding
Content: History of Screening Printing, Advantages, Disadvantages and Applications of screen printing. Screen printing materials: Screen, Frame, Squeegee and Stencil. Screens – multifilament, mono filaments, Selecting mesh material, stretching screen fabric to frame, screen preparation, screen reclamation – Trouble shooting clogged screens. Care and storage of screens. The squeegee, Squeegee considerations, squeegee preparation, hardness categories of squeegee blades, Variety of blade shape and application. Stencils-types, preparation methods of – knife cut stencils, photo stencils – Indirect stencil systems, Direct photo stencil systems, capillary systems, Direct/Indirect photo stencil systems.			
CO2	Explain about substrates and inks used in screen printing and various ink drying methods and related facilities.		
M2.01	Screen-printing substrates	4	Understanding
M2.02	Screen-printing inks	3	Understanding
M2.03	Drying methods	4	Understanding
M2.04	Drying Equipments	3	Understanding
	Series test I	1	

Contents:

Screen-printing substrates: Features & properties of-Paper and Paper board, Wood, Textiles, Plastics, Metals, Ceramics and glass. Screen printing ink-Solvent based, Water based, Radiation Curable. Drying methods - Evaporation, Oxidation, Penetration, and Polymerization. Drying Equipment - Drying racks, wicket dryers, Jet dryers, Infrared dryers, Ultraviolet dryers. Flocking process.

CO3	Explain about on contact & off contact printing, specialized areas, image transfer and registration.		
M3.01	On contact printing, Off contact printing.		Understanding
M3.02	Specialized Areas of screen printing.	3	Understanding
M3.03	Image Transfer and Basic registration techniques.		Understanding
M3.04	Method of halftone preparation for screen printing.	4	Understanding

Contents:

On contact printing, Off contact printing. Specialized Areas - Printed circuit boards of screen printing. Image Transfer and Basic registration techniques. Method of halftone preparation for screen printing.

CO4	Describe about various printing press classifications & configurations of screen-printing.		
M4.01	Manual Printing Process	3	Understanding
M4.02	Semi automatic Screen Printing m/c.	4	Understanding
M4.03	Automatic Screen Printing m/c	4	Understanding
M4.04	Screen Printing machines Configurations	3	Understanding
	Series test II	1	

Contents:

Manual Printing Process, Semi automatic Screen Printing m/c. Automatic Screen Printing m/c. Screen Printing machines Configurations – Flat bet hinged frame, Flat bed vertical lift, Cylinder-bed presses, Container printing m/c, Rotary Screen Printing m/c, Carousel m/c. Special Machine configurations.

Text / Reference:

T/R	Book Title/Author
T1	Adams J.M, Faux,D.D and Rieber L.J, Printing Technology ,5 th Edition, Delmar Publishers, NewYork.
T2	P. Kipphan, Handbook of Print Media , Springer, 2002.
R1	Babette Magee- Screen Printing Preview .
R2	John Stephens.- Screen Printing .

Online resources:

Sl. No	Website Link
1	https://en.wikipedia.org/wiki/Screen_printing
2	https://offbeatpress.com/basics-manual-screen-printing/
3	http://printwiki.org/Screen_Printing
4	https://static1.squarespace.com/static/ScreenprintManual.pdf
5	http://www.spmachine.ru/shelkografiya-en.htm